

Problem 11. Suppose that $\tilde{x}$ is a least squares approximate solution to the linear system $Ax = v$, and suppose that O is an orthogonal matrix. Prove that $\tilde{x}$ is also a least squares approximate solution to $OAx = Ov$.

Solution. To be a least squares approximate solution of $Ax = v$, $\tilde{x}$ must satisfy the condition that $A\tilde{x} - v$ is orthogonal to the columns of A . Because orthogonal matrices preserve orthogonality, $O(A\tilde{x} - v) = OA\tilde{x} - Ov$ is orthogonal to the columns of OA . This means that $\tilde{x}$ is a least squares solution to $OAx = Ov$, for the same reason.

Solution. (Alternative Solution) Because $\tilde{x}$ is a least squares approximate solution to $Ax = v$, it satisfies the normal equation $A^\top A\tilde{x} = A^\top v$. Using the fact that $O^\top O = I_n$, this implies that

$$(OA)^\top OA\tilde{x} = A^\top O^\top OA\tilde{x} = A^\top Ax = A^\top v = A^\top O^\top Ov = (OA)^\top Ov$$

This means that $\tilde{x}$ satisfies the normal equation

$$(OA)^\top OA\tilde{x} = (OA)^\top Ov$$

associated with $OAx = Ov$. This implies that $\tilde{x}$ is also a least squares approximate solution to this equation.

Problem 12. In each part, either find an example of the thing it describes or explain why it is impossible.

(a) An invertible matrix S such that

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = S \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} S^{-1}$$

(b) An orthogonal matrix A such that

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

(c) A 2×4 matrix whose kernel is spanned by the vector

$$\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

(d) Two orthogonal matrices A and B such that $AB \neq BA$.

(e) A matrix A such that $A^2 + 2A$ is the zero matrix, and 3 is an eigenvalue of A .

Solution.

(a) This is impossible, as if it were true, both sides would have the same determinant, but

$$\det \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = -1 \neq 3 = \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \det \left(S \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} S^{-1} \right).$$

(b) This is possible. Such an A satisfies

$$A \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$$

and so we can solve for

$$A = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

which is orthogonal.

(c) This is impossible. If true, we would have $\dim(\ker(A)) = 1$, and $\dim(\operatorname{im}(A)) \leq 2$, but the rank nullity theorem implies that $\dim(\ker(A)) + \dim(\operatorname{im}(A)) = 4$ in this case.

(d) This is possible, with 2×2 matrices. Let A be the reflection

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

and B be the rotation

$$B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = BA.$$

(e) This is impossible. If 3 is an eigenvalue of A , then $Av = 3v$ for some nonzero v . This implies that

$$(A^2 + 2A)v = A(Av) + 2Av = A(3v) + 6v = 9v + 6v = 15v$$

This is nonzero as $v \neq 0$, which contradicts the fact that $A^2 + 2A$ is the zero matrix.